

Course: Computational Sciences 23/24
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INCOMPRESSIBLE NAVIER STOKES EQUATIONS

PROJECT ASSIGNMENTS FOR 5 – 6 STUDENTS

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1. INTRODUCTION

Although a cooking pot might not seem as the prototypical subject of interest for a better understanding of our atmosphere, the basic mechanisms for the generation of weather are quite similar in many regards. More surprisingly the underlying equations pose severe technical and mathematical difficulties, even after more than 150 years of study.

2. MODEL

Consider an incompressible fluid with velocity $\vec{u} = (v(t, y, z), w(t, y, z))$, homogeneous density $\rho > 0$ and viscosity $\nu > 0$. The temperature and the pressure is denoted by $T(t, y, z)$ and $p(t, y, z)$ respectively. Assuming the Boussinesq approximation, the initial value problem for the Navier Stokes equation is given in the following form.

$$\begin{aligned}
 (1) \quad & \partial_t T + \vec{u} \cdot \nabla T = \Delta T \\
 (2) \quad & \partial_t \vec{u} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \nu \Delta \vec{u} + T \beta e_z \\
 (3) \quad & \nabla \cdot \vec{u} = 0 \\
 (4) \quad & \vec{u}(0, y, z) = \vec{u}_0(y, z) \\
 (5) \quad & T(0, y, z) = T_0(y, z)
 \end{aligned}$$

Let us consider the problem on the domain $\Omega = [0, 1]^2$. imposing boundary conditions

$$\begin{aligned}
 \vec{u}(t, 0, z) &= \vec{u}(t, 1, z) \\
 w(t, y, 0) &= 0 = w(t, y, 1) \\
 T(t, y, 0) &= 1 \\
 T(t, y, 1) &= 0 \\
 T(t, 0, z) &= T(t, 1, z),
 \end{aligned}$$

for every $t \in [0, t_{end}]$ and $(y, z) \in \Omega$.

3. TASKS

Tasks marked by * are optional. You need to solve all mandatory assignments. Please use the Wiki of the repository for documentation and answering the questions or provide a L^AT_EX report. The final software should be a python package including a `setup.py`, contain a `Readme.md` explaining the usage as well as a file tracking the dependencies named `requirements.txt`.

3.1. Project management. Apart from the scientific content you should work on the following organizational assignments.

- Setup a development process. For this purpose do research on the matter, document and report your findings. You do not have to use an existing one, but they could serve as a blueprint for your own.
- Maintain a proper git history for each project member in line with the chosen development process.
- * Use GitLab for issue management.
- * Setup and utilize GitLab's continuous integration tools for your tests.

3.2. Introduction. We aim to solve the quite involved equations by splitting the terms into sub problems of easier type. These are then approximated by finite-difference methods on a cartesian grid.

- Explain the phrase *operator splitting*. Why/ How does it work?
- Implement the upwind finite difference method for the linear advection equation with variable advection field in two spatial dimensions using operator splitting for each of the spatial derivatives.

$$\partial_t \vec{u} + \vec{a} \cdot \nabla \vec{u} = 0$$

$$\vec{u}(0, y, z) = \vec{u}_0(y, z)$$

As boundary conditions consider the ones given in the previous section. Check which restrictions occur due to the explicit treatment of this term.

- * Implement a semi-lagrangian scheme for the two dimensional linear advection equation with variable advection field. Such a scheme tracks the characteristic solutions back in time and subsequently interpolates the field at the old time point to obtain the values transported along the characteristics.
- Find suitable testcases, for which you can find analytical solutions and compare them with your results. Therefore visualize the analytical/ approximated solutions and explain what you observe. Numerically test the consistency order of you implementation(s).

3.3. Incompressible flow.

- Familiarize yourself with the concept of Chorin's projection method. Explain the necessity for a projection method.
- Write a Poisson solver in two spatial dimensions, which discretizes the solution of (6) – (7) by finite differences on a cartesian grid.

$$(6) \quad \Delta p = f$$

$$(7) \quad \frac{\partial p}{\partial n} = 0$$

For this purpose define the interface and implement unit tests as well tests to check the consistency first.

- Do the implementation of the solver subsequently.
- Implement Chorin’s projection method [1] for the Navier Stokes equations i.e. (2) – (3) with $\beta = 0$ using the numerical methods built so far. Implement initial conditions to test your code, check the consistency order and visualize the data.

3.4. Rayleigh-Benard convection.

- Implement the scalar equation (1) using the same (explicit) methods as before.
- Implement the source term for $\beta \neq 0$ in (2) by an explicit time stepping method.
- Find initial conditions, boundary conditions and parameters such that you can the classical Rayleigh-Benard convection cells.

3.5. * **Parallelize your code.** We want to make sure to leverage all (or at least more) of the computational resources on your computer hardware efficiently. For this purpose

- Determine and explain the time critical parts in your code by benchmarks. Discuss the different parallelization options fitting to your code. Can you implement all of them in python?

Apply one or more of following strategies. Test and benchmark your code against unoptimized results. Report and explain your findings.

- Parallelize the numerical methods for the advection terms using DASK. What can you do about the Poisson solver?
- Implement a matrix free version of the Poisson solver. Compare the performance to your first implementation.
- Improve performance by offloading the code to accelerators via `pyopencl` or `pycuda`.

REFERENCES

- [1] A. J. Chorin, “The numerical solution of the navier-stokes equations for an incompressible fluid,” *Bull. Amer. Math. Soc.*, vol. 73, pp. 928–931, 11 1967.